SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES SIMATS ENGINEERING

LIST OF EXPERIMENTS

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Day-1

S.NO	LIST OF PROGRAMS	CO
1	Write the python program to solve 8-Puzzle problem	CO1
2	Write the python program to solve 8-Queen problem	CO1
3	Write the python program for Water Jug Problem	CO2
4	Write the python program for Cript-Arithmetic problem	CO2
5	Write the python program for Missionaries Cannibal problem	CO3
6	Write the python program for Vacuum Cleaner problem	CO3
7	Write the python program to implement BFS.	CO3
8	Write the python program to implement DFS.	CO4
9	Write the python to implement Travelling Salesman Problem	CO4
10	Write the python program to implement A* algorithm	CO4

Day-2

S.NO	LIST OF PROGRAMS	CO
1	Write the python program for Map Coloring to implement CSP.	CO5
2	Write the python program for Tic Tac Toe game	CO3
3	Write the python program to implement Minimax algorithm for gaming	CO3
4	Write the python program to implement Apha & Beta pruning algorithm for gaming	CO3
5	Write the python program to implement Decision Tree	CO4
6	Write the python program to implement Feed forward neural Network	CO4
7	Write a Prolog Program to Sum the Integers from 1 to n.	CO4
8	Write a Prolog Program for A DB WITH NAME, DOB.	CO5
9	Write a Prolog Program for STUDENT-TEACHER-SUB-CODE.	CO1
10	Write a Prolog Program for PLANETS DB.	CO2

Day-3

S.NO	LIST OF PROGRAMS	CO
1	Write a Prolog Program to implement Towers of Hanoi.	CO2
2	Write a Prolog Program to print particular bird can fly or not. Incorporate required queries.	CO3
3	Write the prolog program to implement family tree.	CO3
4	Write a Prolog Program to suggest Dieting System based on Disease.	CO3
5	Write a Prolog program to implement Monkey Banana Problem	CO4
6	Write a Prolog Program for fruit and its color using Back Tracking.	CO4
7	Write a Prolog Program to implement Best First Search algorithm	CO4
8	Write the prolog program for Medical Diagnosis	CO4
9	Write a Prolog Program for forward Chaining. Incorporate required queries.	CO5
10	Write a Prolog Program for backward Chaining. Incorporate required queries.	CO1

Day-4

S.NO	LIST OF PROGRAMS	CO
1	Create a Web Blog using Word press to demonstrate Anchor Tag, Title Tag, etc.	CO2
2	Implement a basic linear regression algorithm for predicting a continuous variable.	CO4
3	Write a program to implement the k-nearest neighbors algorithm for classification.	CO5
4	Create a neural network using a popular deep learning library like TensorFlow or PyTorch for image classification.	CO3
5	Implement a decision tree algorithm for a simple classification problem.	CO3
6	Write a program to perform sentiment analysis using Natural Language Processing (NLP) techniques.	CO3
7	Create a genetic algorithm for optimizing a simple mathematical function.	CO4
8	Implement a support vector machine (SVM) for binary classification.	CO4
9	Develop a program to perform dimensionality reduction using Principal Component Analysis (PCA).	CO4
10	Write a code to implement a basic recommendation system using collaborative filtering.	CO5

Day-5

S.NO	LIST OF PROGRAMS	CO

1	Create a simple chatbot using natural language processing techniques.	CO1
2	Implement a clustering algorithm, such as k-means, for grouping similar data points.	CO2
3	Write a program to train a reinforcement learning agent for a basic game or environment.	CO2
4	Develop a program to perform image segmentation using computer vision techniques.	CO3
5	Implement a basic Markov Chain model for predicting sequences of events.	CO3
6	Create a program that uses a pre-trained machine learning model for object detection.	CO3
7	Write a code to perform hyper parameter tuning for a machine learning model.	CO4
8	Implement a Bayesian network for modeling probabilistic relationships between variables.	CO4
9	Develop a program to perform face recognition using a pre-trained model.	CO5
10	Write a genetic algorithm to evolve a neural network for solving a specific problem.	CO1